

Defining Limits

A limit is an intended y value.

As x approaches a certain number, what does the function (the y value) approach?

LIMIT NOTATION

$$\lim_{x \rightarrow 5} x + 3 = 8$$

Reads as

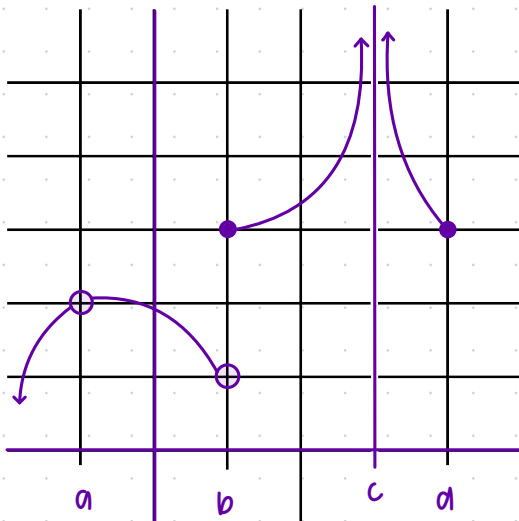
The limit of $x + 3$ as x approaches 5 equals 8

WHAT DOES IT MEAN FOR A LIMIT TO EXIST?

- $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$ - and + indicate whether the function is approaching the x value from the left (-) or the right (+)
- $\lim_{x \rightarrow a} f(x) = L$ (L must be a number) This includes 0, but not $+\infty$ or $-\infty$

DETERMINING WHETHER THE FOLLOWING LIMITS EXISTS USING THE GRAPH BELOW WITH SUPPORTING REASONS

Graph of $f(x)$



$\lim_{x \rightarrow a}$	Exists $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$
$\lim_{x \rightarrow b}$	Does Not Exist (DNE) b/c $\lim_{x \rightarrow b^-} f(x) \neq \lim_{x \rightarrow b^+} f(x)$
$\lim_{x \rightarrow c^-}$	Does Not Exist b/c $\lim_{x \rightarrow c^-} f(x) = \infty$ (Not a #)
$\lim_{x \rightarrow c^+}$	Does Not Exist b/c $\lim_{x \rightarrow c^+} f(x) = \infty$ (Not a #)
$\lim_{x \rightarrow d}$	$\lim_{x \rightarrow d} f(x)$ Does Not Exist only approaches one side